

A Smooth-Submodel Lower Bound for Max-Structured Weakly Convex Optimization

Partial progress toward Zhao's lower-complexity question

June 23, 2026

Abstract

This note gives *partial progress*, not a complete solution, to the lower-complexity question for Zhao's max-structured weakly convex optimization model. We prove that every deterministic method that guarantees an ε -near-stationary point for all instances with weak-convexity parameter γ and initial gap at most Δ must make

$$\Omega\left(\frac{\gamma\Delta}{\varepsilon^2}\right)$$

primal first-order oracle calls in the worst case. In Zhao's more general Bregman normalization, where the residual target is ε/β_X , the same proof yields the sharpened scaling $\Omega(\gamma\Delta\beta_X^2/\varepsilon^2)$. The reduction is from ordinary smooth nonconvex minimization: Zhao's model contains, as a very special submodel, the minimization of a γ -smooth nonconvex function. The result is therefore relevant to the problem in Zhao's paper, but it does not resolve the full question. In particular, the construction gives no lower bound on the dual-gradient complexity and does not explain whether the dependences on L_{xy} , L_{yy} , or the upper-bound parameter L_{xx} in Zhao's upper bounds are necessary.

1 Status of the result

Zhao studies problems of the form

$$q^* = \inf_{x \in \mathcal{X}} q(x), \quad q(x) = f(x) + r(x), \quad f(x) = \max_{y \in \mathcal{Y}} \{\Phi(x, y) - g(y)\}, \quad (1)$$

under smoothness, concavity, and weak-convexity assumptions on Φ ; see [1]. Zhao also distinguishes the primal and dual first-order oracle complexities, because calls returning $\nabla_x \Phi$ and $\nabla_y \Phi$ can have different costs in applications.

The contribution below should be classified as follows.

Classification. The argument is mathematically related to Zhao's lower-complexity question and proves a genuine worst-case lower bound for the same problem class and stationarity notion. It is only partial progress. It proves the unavoidable smooth-nonconvex term $\Omega(\gamma\Delta/\varepsilon^2)$ for primal first-order calls, but it does not determine the full primal-dual complexity of (1).

The proof is intentionally simple. We embed an arbitrary smooth nonconvex function h into (1) by making the dual maximization trivial. Thus any algorithm for (1) would also solve the classical smooth nonconvex stationarity problem. This proves a lower bound that every complete answer must contain, but it leaves open whether there are additional lower-bound terms caused by the genuinely max-structured or primal-dual part of the model.

2 Model and stationarity measure

We work in finite-dimensional Euclidean spaces. This is enough for a lower bound, since Euclidean spaces are a subcase of the finite-dimensional normed spaces in [1]. The sets $\mathcal{X}$ and $\mathcal{Y}$ are nonempty, closed, and convex, and $\mathcal{Y}$ is bounded. The functions r and g are closed, convex, and proper, and have simple proximal projections. For every $y \in \mathcal{Y}$, the map $x \mapsto \Phi(x, y)$ is differentiable and γ -weakly convex on $\mathcal{X}$. For every $x \in \mathcal{X}$, the map $y \mapsto \Phi(x, y)$ is concave on $\mathcal{Y}$. We use the following Lipschitz upper bounds:

$$\|\nabla_x \Phi(x, y) - \nabla_x \Phi(x', y)\| \leq L_{xx} \|x - x'\|, \quad (2)$$

$$\|\nabla_x \Phi(x, y) - \nabla_x \Phi(x, y')\| \leq L_{xy} \|y - y'\|, \quad (3)$$

$$\|\nabla_y \Phi(x, y) - \nabla_y \Phi(x', y)\| \leq L_{xy} \|x - x'\|, \quad (4)$$

$$\|\nabla_y \Phi(x, y) - \nabla_y \Phi(x, y')\| \leq L_{yy} \|y - y'\|. \quad (5)$$

Here $0 < \gamma \leq L_{xx}$ and $L_{xy}, L_{yy} \geq 0$.

We use the Euclidean distance-generating function

$$\omega_X(x) = \frac{1}{2} \|x\|^2, \quad D_{\omega_X}(u, x) = \frac{1}{2} \|u - x\|^2.$$

For $0 < \lambda < 1/\gamma$, define the Moreau-type proximal point

$$\text{prox}(q, x, \lambda) := \arg \min_{u \in \mathcal{X}} \left\{ q(u) + \frac{1}{2\lambda} \|u - x\|^2 \right\}. \quad (6)$$

The minimizer is unique whenever q is γ -weakly convex and $\lambda < 1/\gamma$.

We shall prove the lower bound for the following weak target condition:

$$\lambda^{-1} \|x - \text{prox}(q, x, \lambda)\| \leq \varepsilon \quad \text{for at least one } \lambda \in (0, 1/\gamma). \quad (7)$$

This is no harder than a version that requires (7) for a prescribed λ , or for every admissible λ . Therefore a lower bound for (7) also applies to those stronger target requirements. In Zhao's notation with a general distance-generating function, this is the same residual criterion in the special Euclidean case $\beta_X = 1$.

We count primal first-order oracle calls. A primal oracle call at (x, y) returns $\nabla_x \Phi(x, y)$, and a dual oracle call returns $\nabla_y \Phi(x, y)$. The lower bound below remains valid if dual oracle calls and the proximal projections for r and g are free. It also remains valid for a stronger value-gradient oracle, since the smooth lower bound used below already allows function-value information.

3 A standard smooth nonconvex lower bound

We use the following standard consequence of the high-dimensional lower-bound construction of Carmon, Duchi, Hinder, and Sidford [2]. The exact numerical constants are irrelevant for the reduction.

Lemma 1 (Smooth nonconvex first-order lower bound). *There are universal constants $c_0, c_1 > 0$ such that the following holds. Fix $\gamma > 0$, $\Delta > 0$, and a stationarity tolerance τ satisfying*

$$0 < \tau \leq c_1 \sqrt{\gamma \Delta}.$$

Any deterministic first-order method that, for every differentiable function $h : \mathbb{R}^d \rightarrow \mathbb{R}$ with γ -Lipschitz gradient and initial gap $h(u_0) - \inf h \leq \Delta$, guarantees an output u with

$$\|\nabla h(u)\| \leq \tau$$

requires at least

$$c_0 \frac{\gamma \Delta}{\tau^2}$$

first-order value-gradient oracle queries in the worst case, where the dimension d may depend on the method and on the parameters.

We shall apply Lemma 1 with $\tau = 2\varepsilon$.

4 Embedding smooth minimization into Zhao's model

The following proposition is the whole reduction.

Proposition 1 (Smooth submodel embedding). *Fix parameters*

$$L_{xx} \geq \gamma > 0, \quad L_{xy} \geq 0, \quad L_{yy} \geq 0.$$

Let $h : \mathbb{R}^d \rightarrow \mathbb{R}$ be continuously differentiable, bounded below, and have γ -Lipschitz gradient. Define the primal variable

$$x = (u, w, z) \in \mathbb{R}^d \times \mathbb{R} \times \mathbb{R},$$

the feasible sets

$$\mathcal{X} := \mathbb{R}^d \times \mathbb{R} \times \{0\}, \quad \mathcal{Y} := [-1, 1],$$

the simple convex functions

$$r := \iota_{\mathcal{X}}, \quad g \equiv 0,$$

and

$$\Phi((u, w, z), y) := h(u) + \frac{L_{xx}}{2}w^2 + L_{xy}zy - \frac{L_{yy}}{2}y^2. \quad (8)$$

Then the instance belongs to the model (1) with upper bounds L_{xx}, L_{xy}, L_{yy} and weak-convexity parameter γ . Moreover, on the feasible set $z = 0$,

$$q(u, w, 0) = h(u) + \frac{L_{xx}}{2}w^2. \quad (9)$$

Consequently, if $x_0 = (u_0, 0, 0)$, then

$$q(x_0) - q^* = h(u_0) - \inf h.$$

Finally, if $x = (u, w, 0) \in \mathcal{X}$ satisfies the near-stationarity condition (7) for the constructed instance, then

$$\|\nabla h(u)\| < 2\varepsilon.$$

Proof. First observe that $\mathcal{Y} = [-1, 1]$ is compact and convex, r and g are closed convex proper functions, and r is identically zero on $\mathcal{X}$. The map $y \mapsto \Phi(x, y)$ is concave because it is affine minus $(L_{yy}/2)y^2$.

The gradients are

$$\begin{aligned}\nabla_x \Phi((u, w, z), y) &= (\nabla h(u), L_{xx}w, L_{xy}y), \\ \nabla_y \Phi((u, w, z), y) &= L_{xy}z - L_{yy}y.\end{aligned}$$

For fixed y ,

$$\begin{aligned}\|\nabla_x \Phi((u, w, z), y) - \nabla_x \Phi((u', w', z'), y)\|^2 &= \|\nabla h(u) - \nabla h(u')\|^2 + L_{xx}^2 |w - w'|^2 \\ &\leq L_{xx}^2 (\|u - u'\|^2 + |w - w'|^2 + |z - z'|^2),\end{aligned}$$

because ∇h is γ -Lipschitz and $\gamma \leq L_{xx}$. This gives (2). Similarly,

$$\|\nabla_x \Phi(x, y) - \nabla_x \Phi(x, y')\| = L_{xy}|y - y'|,$$

so (3) holds. Also,

$$|\nabla_y \Phi((u, w, z), y) - \nabla_y \Phi((u', w', z'), y)| = L_{xy}|z - z'| \leq L_{xy} \|x - x'\|,$$

and

$$|\nabla_y \Phi(x, y) - \nabla_y \Phi(x, y')| = L_{yy}|y - y'|,$$

which prove (4) and (5).

For weak convexity, use the standard fact that γ -smooth differentiability implies γ -weak convexity:

$$h(v) \geq h(u) + \langle \nabla h(u), v - u \rangle - \frac{\gamma}{2} \|v - u\|^2.$$

The term $(L_{xx}/2)w^2 + L_{xy}zy$ is convex in (w, z) up to its affine part in z on the affine feasible set $\mathcal{X}$, where $z = 0$. More directly, restricted to $\mathcal{X}$, the only nonconstant primal terms are $h(u)$ and $(L_{xx}/2)w^2$, and the latter is convex. Hence $x \mapsto \Phi(x, y)$ is γ -weakly convex on $\mathcal{X}$ for every $y \in \mathcal{Y}$.

On $\mathcal{X}$ we have $z = 0$, so

$$\Phi((u, w, 0), y) = h(u) + \frac{L_{xx}}{2}w^2 - \frac{L_{yy}}{2}y^2.$$

The maximum over $y \in [-1, 1]$ equals the value at $y = 0$, namely (9). Thus $q^* = \inf h$, attained in the w -coordinate at $w = 0$ if the infimum of h is approached.

It remains to prove the stationarity transfer. Let

$$x^+ := \text{prox}(q, x, \lambda) = (u^+, w^+, 0)$$

for a $\lambda \in (0, 1/\gamma)$ satisfying (7). Because of the separable formula (9), the u -component of the proximal point is

$$u^+ = \arg \min_{v \in \mathbb{R}^d} \left\{ h(v) + \frac{1}{2\lambda} \|v - u\|^2 \right\}. \quad (10)$$

The objective in (10) is strongly convex, since h is γ -weakly convex and $1/\lambda > \gamma$. From (7),

$$\lambda^{-1} \|u - u^+\| \leq \lambda^{-1} \|x - x^+\| \leq \varepsilon.$$

The first-order optimality condition for (10) gives

$$\nabla h(u^+) + \lambda^{-1}(u^+ - u) = 0,$$

so $\|\nabla h(u^+)\| \leq \varepsilon$. Finally, using the γ -Lipschitz continuity of ∇h ,

$$\|\nabla h(u)\| \leq \|\nabla h(u^+)\| + \gamma \|u - u^+\| \leq \varepsilon + \gamma \lambda \varepsilon < 2\varepsilon,$$

because $\lambda < 1/\gamma$. □

5 Main lower bound

Theorem 1 (Primal first-order lower bound). *Fix*

$$L_{xx} \geq \gamma > 0, \quad L_{xy} \geq 0, \quad L_{yy} \geq 0, \quad \Delta > 0.$$

There are universal constants $c, C > 0$ such that, for every

$$0 < \varepsilon \leq C\sqrt{\gamma\Delta},$$

any deterministic method that guarantees an ε -near-stationary point, in the sense of (7), for every instance of (1) satisfying the assumptions above and having initial gap at most Δ , must make at least

$$c \frac{\gamma\Delta}{\varepsilon^2} \tag{11}$$

primal first-order oracle calls in the worst case. The statement remains true if dual oracle calls, the set projections, and the proximal projections associated with r and g are free.

Proof. Set $\tau = 2\varepsilon$ and choose $C \leq c_1/2$, where c_1 is the constant in Lemma 1. Suppose, for contradiction, that an algorithm A solves all such max-structured instances using fewer than

$$\frac{c_0}{4} \frac{\gamma\Delta}{\varepsilon^2} = c_0 \frac{\gamma\Delta}{\tau^2}$$

primal oracle calls.

We construct a first-order method B for the smooth nonconvex problem. Given a hard instance h and initial point u_0 from Lemma 1, method B forms the instance in Proposition 1 and runs A on it from $x_0 = (u_0, 0, 0)$.

Whenever A requests a primal oracle call at $((u, w, z), y)$, method B makes one value-gradient query to h at u and returns

$$\nabla_x \Phi((u, w, z), y) = (\nabla h(u), L_{xx}w, L_{xy}y),$$

along with the value of Φ if the oracle model includes values. All terms except $h(u)$ and $\nabla h(u)$ are explicit. Whenever A requests a dual oracle call, B returns

$$\nabla_y \Phi((u, w, z), y) = L_{xy}z - L_{yy}y,$$

which requires no query to h . The proximal operations for r and g are also explicit. Thus the number of first-order queries made by B to h is exactly the number of primal oracle calls made by A.

By Proposition 1, the constructed max-structured instance satisfies all parameter bounds and has initial gap at most Δ . Therefore A outputs a point $x = (u, w, 0)$ satisfying (7). Proposition 1 then gives

$$\|\nabla h(u)\| < 2\varepsilon = \tau.$$

So B finds a τ -stationary point of every smooth hard instance using fewer than $c_0\gamma\Delta/\tau^2$ first-order queries, contradicting Lemma 1. This proves (11) with $c = c_0/4$. $\square$

Remark 1 (Bregman normalization and the factor β_X). *The theorem was stated in the Euclidean normalization $\beta_X = 1$ to keep the main proof uncluttered. The same reduction gives the corresponding bound for Zhao's Bregman residual convention. Namely, replace the Euclidean distance-generating function by*

$$\omega_X(u, w, z) = \frac{1}{2} \|u\|^2 + \frac{\beta_X}{2} w^2 + \frac{1}{2} z^2, \quad \beta_X \geq 1.$$

Restricted to $\mathcal{X} = \mathbb{R}^d \times \mathbb{R} \times \{0\}$, this function is 1-strongly convex and has β_X -Lipschitz gradient. The u -component of the proximal point is still the Euclidean proximal point

$$u^+ = \arg \min_{v \in \mathbb{R}^d} \left\{ h(v) + \frac{1}{2\lambda} \|v - u\|^2 \right\}.$$

If the near-stationarity criterion is

$$\lambda^{-1} \|x - \text{prox}(q, x, \lambda)\| \leq \varepsilon/\beta_X,$$

then the proof of Proposition 1 gives

$$\|\nabla h(u)\| < 2\varepsilon/\beta_X.$$

Applying Lemma 1 with tolerance $\tau = 2\varepsilon/\beta_X$ yields the lower bound

$$\Omega\left(\frac{\gamma\Delta\beta_X^2}{\varepsilon^2}\right),$$

under the same harmless small-tolerance condition $\varepsilon/\beta_X = O(\sqrt{\gamma\Delta})$.

Corollary 1 (Singleton-dual subcase). *If $\mathcal{Y}$ is a singleton, $r \equiv 0$, $g \equiv 0$, and $\Phi(x, y) = h(x)$, Zhao's model reduces to ordinary smooth nonconvex minimization. Consequently, even the subclass with $L_{xx} = \gamma$ and $L_{xy} = L_{yy} = 0$ already has worst-case primal first-order complexity*

$$\Omega\left(\frac{L_{xx}\Delta}{\varepsilon^2}\right).$$

6 What this does and does not settle

Theorem 1 is a genuine lower bound for the max-structured weakly convex class. It rules out any deterministic method whose primal first-order complexity is $o(\gamma\Delta/\varepsilon^2)$ uniformly over the class. In that sense it makes actual progress on the lower-complexity question.

However, it is not a complete solution of Zhao's question. The proof deliberately makes the dual side trivial. Therefore it cannot prove any nonzero lower bound on the number of dual-gradient calls: in the hard instances above, $\nabla_y \Phi$ is explicit and carries no information about the hidden smooth function h . The proof also does not show that positive values of L_{xy} or L_{yy} make the problem harder. Those parameters are present only through dummy coordinates that allow the instance to satisfy prescribed upper bounds.

Thus the theorem should be interpreted as the baseline lower-bound term forced by the smooth nonconvex submodel:

Any complete lower-complexity theory must contain $\Omega(\gamma\Delta/\varepsilon^2)$ primal calls.

What remains open, after this reduction, is whether genuinely primal-dual hard instances imply additional terms depending on L_{xy} , L_{yy} , L_{xx} , the geometry of $\mathcal{Y}$, or the distinction between primal and dual oracle costs.

Remark 2 (On the gap parameter). *The dependence on the initial gap $\Delta = q(x_0) - q^*$ is not an artifact of the proof. Smooth nonconvex hard families can be concatenated so that a method must traverse many hidden components, and the number of components is controlled by the allowed total objective decrease. Consequently, a dimension-free worst-case lower bound for stationarity requires a scale parameter such as Δ .*

References

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